

Differential Gene-Expression Analysis — Healthy vs Disease (Inflammation Panel)

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Objective

To identify genes whose expression differs significantly between Healthy and Disease groups in a six-gene inflammation/regeneration panel (IL6, TNF, IL1B, IL10, TGFB1, VEGFA), and to demonstrate a reproducible differential-expression workflow with multiple-testing correction.

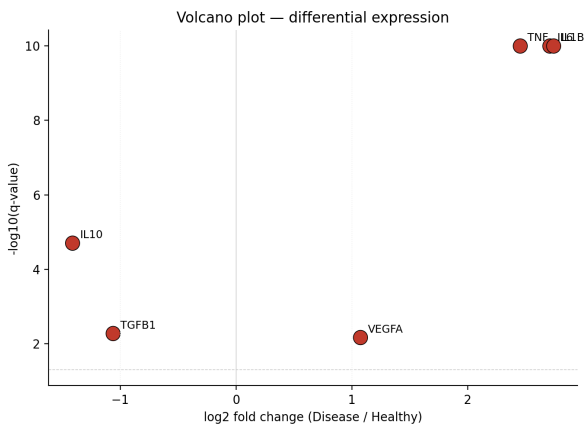
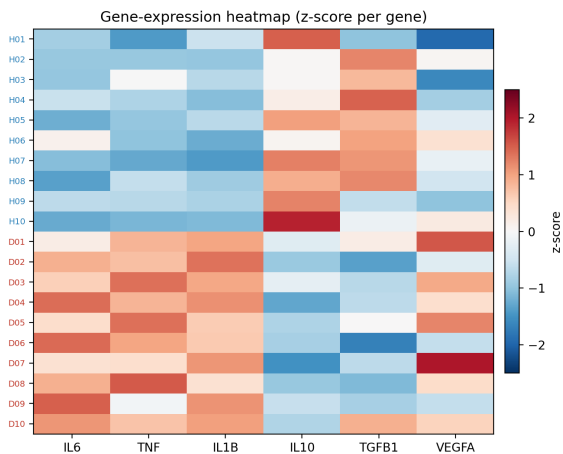
Methods

Simulated log2-normalised expression for 20 samples (10 Healthy, 10 Disease) was analysed in Python (pandas, SciPy, matplotlib). For each gene I computed group means, log2 fold change (Disease vs Healthy), and a Welch two-sample t-test. The Benjamini-Hochberg procedure was applied across the six genes to control the false-discovery rate (q-value). Genes were flagged differentially expressed at $|\log_2FC| > 1$ and $q < 0.05$.

Results

Gene	Healthy mean	Disease mean	log2 FC	p-value	q-value	Sig (q<0.05)
IL10	5.58	4.17	-1.41	1.33e-05	2.0e-05	Yes
IL1B	4.23	6.97	+2.74	1.69e-11	0.0e+00	Yes
IL6	4.20	6.91	+2.71	7.18e-08	0.0e+00	Yes
TGFB1	6.06	5.00	-1.06	0.00449	5.4e-03	Yes
TNF	4.30	6.75	+2.45	6.58e-08	0.0e+00	Yes
VEGFA	5.12	6.19	+1.07	0.0069	6.9e-03	Yes

Upregulated in Disease ($|\log_2FC| > 1$, $q < 0.05$): IL1B, IL6, TNF, VEGFA. **Downregulated:** IL10, TGFB1. **Below threshold:** none.



Conclusion

The pro-inflammatory cytokines IL6, TNF, and IL1B were significantly upregulated in the Disease group, while the anti-inflammatory cytokine IL10 was significantly downregulated — a pattern consistent with a shift towards a pro-inflammatory state. VEGFA was upregulated, suggesting compensatory angiogenic activity. The workflow (data → fold change → t-test → FDR correction → heatmap + volcano) mirrors a real differential-expression analysis at the smallest possible scale and is fully reproducible from a single command.